



FIRST REVIEW – PROJECT WORK (PRIR19)

Project Title: Detection and Analysis of Silent Decision Degradation in AI-Assisted Inventory Decision Systems

1. Problem Statement & Objectives

AI-driven inventory and logistics systems are typically evaluated using static performance metrics. In dynamic, partially observable environments, such systems may continue operating while their decision quality degrades silently. This project studies how and why such degradation occurs, and how it can be detected before operational failure.

Project Objectives

Objective ID	Objective Description
O1	Model a realistic inventory allocation decision system
O2	Establish baseline decision performance under stable assumptions
O3	Inject controlled failures to violate system assumptions
O4	Measure gradual degradation beyond traditional metrics
O5	Prepare ground for detection and recovery mechanisms

2. Methodology / Approach

Overall Methodology Structure

Phase	Name	Purpose
Phase 0	Environment Characterization	Understand data structure, coupling, volatility
Phase 1	Baseline Decision System	Create a reference decision model
Phase 2	Failure Injection	Induce silent degradation
Phase 3	Degradation Detection	Identify early warning signals
Phase 4	Correction & Recovery	Stabilize degraded decisions

Phase-wise Technical Approach

Phase	Key Activities	Outputs
Phase 0	EDA, network analysis, demand profiling	Structural understanding
Phase 1	Inventory allocation logic, metric design	Baseline performance
Phase 2	Demand drift, delays, coupling	Degraded behavior
Phase 3	Behavioral & structural indicators	Detection signals
Phase 4	Adaptive correction strategies	Recovery validation

Horizontal & Vertical Expansion Strategy

Expansion Type	Description
Horizontal	Multiple decision contexts, observation regimes
Vertical	Increasing depth of failure and detection complexity

3. Work Completed So Far

Completed Phases

Phase	Status	Remarks
Phase 0	Completed	Environment fully characterized
Phase 1	Completed	Baseline system implemented
Phase 2	Completed	9 Failures modes successfully injected

Implemented Components:

Component	Status
Data integration & cleaning	Done
Exploratory data analysis	Done
Network-based structural analysis	Done
Baseline inventory allocation model	Done
Performance metric computation	Done

Failure injection framework	Done
Signal failure experiments (3 modes)	Done
Structural failure experiments (3 modes)	Done
Behavioral failure experiments (3 modes)	Done
Streamlit monitoring dashboard	Done

4. Preliminary Results / Observations**Key Observations**

Aspect	Observation
Demand	Strong seasonality and inter-store variability
Holidays	Demand amplification with significant variance
Allocation	High service level masks inefficiencies
Metrics	Aggregate metrics hide degradation patterns
Structure	Strong inter-store and inter-department coupling
Behavior	Feedback mechanisms introduce allocation oscillations

Baseline Performance Summary

Metric	Value(Observed)
Average Service Level	~0.92
Total Unmet Demand	High
Total Overstock	High
Allocation Volatility	High
Economic Loss Exposure	Significant

5. Future Plan & Timeline**Planned Activities**

Timeline	Activity
Mar–Apr	Degradation signal design (Phase 3)
Apr	Recovery mechanisms (Phase 4)
May	Final analysis & report

6. Other Details

Item	Description
Dataset	Walmart Store Sales (public)
Tools	Python, Jupyter, NetworkX
Focus	Decision behavior, not model accuracy
Output	Research-oriented FYP

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Date: 17/03/2026

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